

Statistics and Probability  
Assignment - 01

SPRIHA PODDER  
23FE10CSE00471  
(SE (core) - Sec K

- Q1. (a) A random variable is a function that associates a real number with each element in a sample space.
- (b) If two random variables are independent, then knowing the value of one of them does not change the probabilities for the other one.
- (c) According to Chebyshev's inequality, if  $X$  is a random variable with mean  $\mu$  and variance  $\sigma^2$ , then for any positive number  $K$ , we can say,
- $$P\{|X - \mu| \geq K\sigma\} \leq \frac{1}{K^2}$$
- OR.  $P\{|X - \mu| \leq K\sigma\} \geq 1 - \frac{1}{K^2}$
- (d) The distribution function describes the dependence between a random variable and its probabilities, i.e., it is a mathematical expression that describes the probability that a system will take on a specific value or a set of values.
- (e) 
$$f(x) = \begin{cases} e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$
- $$\begin{aligned} P(X=x) &= \int_0^{\infty} e^{-x} dx + \int_{-\infty}^0 0 \cdot dx \\ &= [-e^{-x}]_0^{\infty} \\ &= -0 - (-e^0) \\ &= -0 - (-1) \\ &= 1 \end{aligned}$$
- $\therefore \boxed{P(X=x) = 1}$
- Therefore it is a probability density function.
- (f) A discrete random variable can take finite number of values within its range.
- (g) The range of a distribution function is the difference between the maximum and the minimum value.
- (h) (i)  $E(X) \leq E(Y)$
- (i) The height of persons in a country is a random variable of the type discrete.

Q2.  $S = \{ (1,1) (1,2) (1,3) (1,4) (1,5) (1,6)$   
 $(2,1) (2,2) (2,3) (2,4) (2,5) (2,6)$   
 $(3,1) (3,2) (3,3) (3,4) (3,5) (3,6)$   
 $(4,1) (4,2) (4,3) (4,4) (4,5) (4,6)$   
 $(5,1) (5,2) (5,3) (5,4) (5,5) (5,6)$   
 $(6,1) (6,2) (6,3) (6,4) (6,5) (6,6) \}$

|        |                |                |                |                |                |                |                |                |                |                |                |
|--------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| $x$    | 2              | 3              | 4              | 5              | 6              | 7              | 8              | 9              | 10             | 11             | 12             |
| $P(X)$ | $\frac{1}{36}$ | $\frac{2}{36}$ | $\frac{3}{36}$ | $\frac{4}{36}$ | $\frac{5}{36}$ | $\frac{6}{36}$ | $\frac{5}{36}$ | $\frac{4}{36}$ | $\frac{3}{36}$ | $\frac{2}{36}$ | $\frac{1}{36}$ |

$$\begin{aligned} \sum P(X=x_i) &= P(X=1) + P(X=2) + \dots + P(X=12) \\ &= \frac{1+2+3+4+5+6+5+4+3+2+1}{36} \end{aligned}$$

$$= 1$$

$$f(x) = \begin{cases} 0 & , X < 0 \\ 1/36 & , x = 1 \\ 2/36 & , x = 2 \\ 3/36 & , x = 3 \\ 4/36 & , x = 4 \\ 5/36 & , x = 5 \\ 6/36 & , x = 6 \\ 5/36 & , x = 7 \\ 4/36 & , x = 8 \\ 3/36 & , x = 9 \\ 2/36 & , x = 10 \\ 1/36 & , x = 11 \\ 0 & , x > 12 \end{cases}$$

Cumulative Distribution Function

|        |                |                |                |                 |                 |                 |                 |                 |                 |                 |                |
|--------|----------------|----------------|----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|
| $x$    | 2              | 3              | 4              | 5               | 6               | 7               | 8               | 9               | 10              | 11              | 12             |
| $P(X)$ | $\frac{1}{36}$ | $\frac{2}{36}$ | $\frac{3}{36}$ | $\frac{4}{36}$  | $\frac{5}{36}$  | $\frac{6}{36}$  | $\frac{5}{36}$  | $\frac{4}{36}$  | $\frac{3}{36}$  | $\frac{2}{36}$  | $\frac{1}{36}$ |
| $F(x)$ | $\frac{1}{36}$ | $\frac{3}{36}$ | $\frac{6}{36}$ | $\frac{10}{36}$ | $\frac{15}{36}$ | $\frac{21}{36}$ | $\frac{26}{36}$ | $\frac{30}{36}$ | $\frac{33}{36}$ | $\frac{35}{36}$ | 1              |

Q3.  $f(x) = 6x(1-x)$  ,  $0 \leq x \leq 1$

$$\begin{aligned}
 \text{(i)} \quad P(X \leq 1) &= \int_0^1 f(x) dx \\
 &= \int_0^1 [6x - 6x^2] dx \\
 &= \left[ \frac{6x^2}{2} \right]_0^1 - \left[ \frac{6x^3}{3} \right]_0^1 \\
 &= 3(1-0) - 2(1-0) \\
 &= 3-2 \\
 &= 1
 \end{aligned}$$

$\therefore f(x)$  is a valid pdf.

(ii)  $P(X \leq b) = P(X > b)$

$$\begin{aligned}
 P(X \leq b) &= \int_0^b 6x(1-x) dx \\
 &= \left[ 3x^2 \right]_0^b - \left[ 2x^3 \right]_0^b \\
 &= 3b^2 - 2b^3 \\
 &= b^2 [3 - 2b]
 \end{aligned}$$

$$\begin{aligned}
 P(X > b) &= \int_b^1 6x(1-x) dx \\
 &= \left[ 3x^2 \right]_b^1 - \left[ 2x^3 \right]_b^1 \\
 &= 3(1-b^2) - 2(1-b^3) \\
 &= 3 - 3b^2 - 2 + 2b^3 \\
 &= 1 + 2b^3 - 3b^2
 \end{aligned}$$

$$\begin{aligned}
 P(X \leq b) &= P(X > b) \\
 3b^2 - 2b^3 &= 1 + 2b^3 - 3b^2 \\
 6b^2 - 4b^3 - 1 &= 0 \\
 4b^3 - 6b^2 + 1 &= 0 \\
 \therefore \boxed{b = 0.5}
 \end{aligned}$$

(iii)  $x < 0$  ,  $F(x) = 0$

$$\begin{aligned}
 0 \leq x \leq 1, \quad F(x) &= \int_0^x 6x(1-x) dx \\
 &= \left[ 3x^2 \right]_0^x - \left[ 2x^3 \right]_0^x \\
 &= 3x^2 - 2x^3
 \end{aligned}$$

$$F(x) = \begin{cases} 0 & , x < 0 \\ x^2(3-2x) & , 0 \leq x \leq 1 \\ 1 & , x > 1 \end{cases}$$

Cumulative  
Distribution  
Function

$$(iv) P\left(X \leq \frac{1}{2} \mid \frac{1}{3} \leq X \leq \frac{2}{3}\right)$$

$$\begin{aligned} P\left(X \leq \frac{1}{2}\right) &= \int_0^{\frac{1}{2}} [6x - 6x^2] dx \\ &= [3x^2]_0^{\frac{1}{2}} - [2x^3]_0^{\frac{1}{2}} \\ &= \frac{3}{4} - \frac{1}{4} \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} P\left(\frac{1}{3} \leq X \leq \frac{2}{3}\right) &= F\left(\frac{2}{3}\right) - F\left(\frac{1}{3}\right) \\ &= [x^2(3-2x)] - [x^2(3-2x)] \\ &= \frac{4}{9}\left(3-2\left(\frac{2}{3}\right)\right) - \frac{1}{9}\left(3-2\left(\frac{1}{3}\right)\right) \\ &= \frac{4}{9}\left(\frac{5}{3}\right) - \frac{1}{9}\left(\frac{7}{3}\right) \\ &= \frac{20}{27} - \frac{7}{27} \\ &= \frac{13}{27} \end{aligned}$$

$$\begin{aligned} P\left(X \leq \frac{1}{2} \cap \frac{1}{3} \leq X \leq \frac{2}{3}\right) &= P\left(\frac{1}{3} \leq X \leq \frac{1}{2}\right) \\ &= F\left(\frac{1}{2}\right) - F\left(\frac{1}{3}\right) \\ &= \frac{1}{4}\left(3-2\left(\frac{1}{2}\right)\right) - \frac{7}{27} \\ &= \frac{1}{4}(2) - \frac{7}{27} \\ &= \frac{40}{54} - \frac{14}{54} \\ &= \frac{26}{54} \\ &= \frac{13}{27} \end{aligned}$$

$$\begin{aligned} P\left(X \leq \frac{1}{2} \mid \frac{1}{3} \leq X \leq \frac{2}{3}\right) &= \frac{P\left(\frac{1}{3} \leq X \leq \frac{1}{2}\right)}{P\left(\frac{1}{3} \leq X \leq \frac{2}{3}\right)} \\ &= \frac{\frac{13}{27}}{\frac{13}{27}} \\ &= \boxed{\frac{1}{2}} \end{aligned}$$

Q4.

|          |       |       |       |
|----------|-------|-------|-------|
| $x$      | -3    | 6     | 7     |
| $P(X=x)$ | $1/6$ | $1/2$ | $1/3$ |

$$\begin{aligned}
 E(X) &= \sum x_i P(x_i) \\
 &= -3\left(\frac{1}{6}\right) + 6\left(\frac{1}{2}\right) + 7\left(\frac{1}{3}\right) \\
 &= -\frac{1}{2} + 3 + 3 \\
 &= \frac{-1 + 12}{2} \\
 &= \boxed{\frac{11}{2}}
 \end{aligned}$$

$$\begin{aligned}
 E(X^2) &= \sum x_i^2 P(x_i) \\
 &= 9\left(\frac{1}{6}\right) + 36\left(\frac{1}{2}\right) + 49\left(\frac{1}{3}\right) \\
 &= \frac{3}{2} + 18 + 27 \\
 &= \frac{3 + 36 + 54}{2} \\
 &= \boxed{\frac{93}{2}}
 \end{aligned}$$

$$\begin{aligned}
 E(2x+1)^2 &= \sum (2x_i+1)^2 P(x_i) \\
 &= \cancel{-5\left(\frac{1}{6}\right) + 13\left(\frac{1}{2}\right) + 19\left(\frac{1}{3}\right)}
 \end{aligned}$$

$$\begin{aligned}
 &= \cancel{-\frac{5}{6} + \frac{13}{2} + \frac{19}{3}} \\
 &= \cancel{\frac{-5 + 39 + 38}{6}} \\
 &= \cancel{\frac{72}{6}} \\
 &= \boxed{12}
 \end{aligned}$$

$$\begin{aligned}
 &= E(4x^2 + 1 + 4x) \\
 &= 4E(X^2) + 4E(X) + 1 \\
 &= 4\left(\frac{93}{2}\right) + 4\left(\frac{11}{2}\right) + 1 \\
 &= 186 + 22 + 1 \\
 &= \boxed{209}
 \end{aligned}$$

$$Q5. \quad P(-2 < X < 8) \geq \frac{21}{25}$$

$$P(-K\sigma < X - \mu < K\sigma) \geq 1 - \frac{1}{K^2}$$

$$P(\mu - K\sigma < X < \mu + K\sigma) \geq 1 - \frac{1}{K^2}$$

$$\mu - K\sigma = -2$$

——— ①

$$\mu + K\sigma = 8$$

——— ②

Adding ① and ②

$$\mu - K\sigma + \mu + K\sigma = -2 + 8$$

$$2\mu = 6$$

$$\boxed{\mu = 3}$$

Subtracting ① and ②

$$\mu - K\sigma - \mu - K\sigma = -2 - 8$$

$$-2K\sigma = -10$$

$$K\sigma = 5 \quad \text{——— ③}$$

Since,  $\frac{21}{25} = 1 - \frac{1}{K^2}$

$$\frac{1}{K^2} = 1 - \frac{21}{25}$$

$$\frac{1}{K^2} = \frac{4}{25}$$

$$K^2 = \frac{25}{4}$$

$$\boxed{K = \frac{5}{2}}$$

from ③

$$K\sigma = 5$$

$$\frac{5}{2}\sigma = 5$$

$$\therefore \boxed{\sigma = 2}$$

Therefore,  $E(X) = \mu = 3$   
 $\text{Var}(X) = \sigma^2 = 4$

Q6. (i)  $\sum p(x) = 1$   
 $10K^2 + 9K = 1$   
 $10K^2 + 9K - 1 = 0$   
 $10K^2 + 10K - K - 1 = 0$   
 $10K(K+1) - 1(K+1) = 0$   
 $(10K-1)(K+1) = 0$   $\therefore K > 0$   
 $\therefore \boxed{K = \frac{1}{10}}$

(ii)  $P(X < 6) = P(0) + P(1) + P(2) + P(3) + P(4) + P(5)$   
 $= K^2 + 8K = \frac{81}{100} = 0.81$

$P(X = 6) = P(6) + P(7)$   
 $= 9K^2 + K = \frac{19}{100} = 0.19$

$P(0 < X < 5) = P(1) + P(2) + P(3) + P(4)$   
 $= 8K = \frac{8}{10} = 0.8$

(iii)

| $x$ | $P(X \leq x)$ |
|-----|---------------|
| 0   | 0             |
| 1   | 0.1           |
| 2   | 0.3           |
| 3   | 0.5           |
| 4   | 0.8           |
| 5   | 0.81          |
| 6   | 0.83          |
| 7   | 1             |

(iv)  $P(X \leq 0) = 0$   
 $P(X \leq 1) = 0 + K = \frac{1}{10} < \frac{1}{2}$   
 $P(X \leq 2) = 0 + 3K = \frac{3}{10} < \frac{1}{2}$   
 $P(X \leq 3) = 5K = \frac{1}{2}$   
 $P(X \leq 4) = 8K = \frac{8}{10} > \frac{1}{2} \rightarrow \therefore \text{Minimum value of } \boxed{a=4}$

$$Q7. f(x) = kx^2(1-x^3), \quad 0 < x < 1$$

$$P(X \leq x) = 1$$

$$\int_0^1 f(x) dx = 1$$

$$\int_0^1 (kx^2 - kx^5) dx = 1$$

$$\left[ k \frac{x^3}{3} \right]_0^1 - \left[ \frac{kx^6}{6} \right]_0^1 = 1$$

$$\frac{k}{3} - \frac{k}{6} = 1$$

$$\frac{2k - k}{6} = 1$$

$$\boxed{k = 6}$$

$$y = \int_0^1 x \cdot f(x) dx$$

$$= \int_0^1 6x^3(1-x^3) dx$$

$$= \left[ \frac{6x^4}{4} \right]_0^1 - \left[ \frac{6x^7}{7} \right]_0^1$$

$$= \frac{3}{2} - \frac{6}{7}$$

$$= \frac{21-12}{14}$$

$$\boxed{y = \frac{9}{14}}$$

$$y^2 = \int_0^1 x^2 f(x) dx - y^2$$

$$= \int_0^1 6x^4(1-x^3) dx - \left( \frac{9}{14} \right)^2$$

$$= \left( \frac{6x^5}{5} \right)_0^1 - \left[ \frac{6x^8}{8} \right]_0^1 - \left( \frac{9}{14} \right)^2$$

$$= \frac{6}{5} - \frac{6}{8} - \frac{81}{196}$$

$$= \frac{18}{40} - \frac{81}{196}$$

$$= 0.45 - 0.41$$

$$\boxed{y^2 = 0.04}$$

Q8.  $p(x) = e^{-t}(1-e^{-t})^{x-1}$ ,  $x=1,2,3,\dots$

$$\begin{aligned} \int_1^{\infty} p(x) dx &= \int_1^{\infty} e^{-t}(1-e^{-t})^{x-1} dx \\ &= \int_1^{\infty} e^{-t} - \end{aligned}$$

$$p(x=1) = e^{-t}(1-e^{-t})^{1-1} = e^{-t}(1) = e^{-t} > 0$$

$$p(x=2) = e^{-t}(1-e^{-t})^{2-1} = e^{-t}(1-e^{-t}) = e^{-t} - (e^{-t})^2 > 0$$

$$\begin{aligned} p(x=3) &= e^{-t}(1-e^{-t})^{3-1} = e^{-t}(1-e^{-t})^2 \\ &= e^{-t}(1-e^{-t})^2 = e^{-t}(1+e^{-2t}-2e^{-t}) \\ &= e^{-t} + e^{-3t} - 2e^{-2t} > 0 \end{aligned}$$

Since,  $e^{-t} > e^{-2t} > e^{-3t}$

$$\begin{aligned} \sum p(x=x) &= p(x=1) + p(x=2) + p(x=3) \\ &= e^{-t} + e^{-t} - e^{-2t} + e^{-t} + e^{-3t} - 2e^{-2t} \\ &= 3e^{-t} + e^{-3t} - 3e^{-2t} > 0 \end{aligned}$$

$$\begin{aligned} \sum p(x=x) &= \sum_{x=1}^{\infty} e^{-t}(1-e^{-t})^{x-1} \\ &= e^{-t} \sum_{x=0}^{\infty} (1-e^{-t})^x \end{aligned}$$

Since, GP:  $a=1$ ,  $r=1-e^{-t}$

$$S = \frac{a}{1-r} = e^{-t} \cdot \frac{1}{1-(1-e^{-t})} = \boxed{1}$$

$$E(x) = \sum_{x=1}^{\infty} x \cdot p(x) = e^{-t} \sum_{x=1}^{\infty} x \cdot (1-e^{-t})^{x-1}$$

$$= e^{-t} \sum_{x=1}^{\infty} x \cdot (1-e^{-t})^{x-1}$$

$$= e^{-t} \left[ \frac{1 - (1-e^{-t}) + 1 \cdot (1-e^{-t})}{(1-(1-e^{-t}))^2} \right]$$

$$= e^{-t} \cdot \frac{1}{e^{-2t}}$$

$$= \boxed{e^t}$$

AGP  $\Rightarrow$   
 $S = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$

$$E(x^2) = \sum_{x=1}^{\infty} x^2 \cdot p(x) = \sum_{x=1}^{\infty} x^2 \cdot e^{-t}(1-e^{-t})^{x-1} = e^t + 2e^{2t}$$

$$\begin{aligned} \text{Var}(x) &= E(x^2) - [E(x)]^2 \\ &= e^t + 2e^{2t} - e^{2t} \\ &= \boxed{e^t + e^{2t}} \end{aligned}$$

Q9.

$$E(X) = 2$$

$$\sigma_x^2 = 1$$

$$E(Y) = 3$$

$$\sigma_y^2 = 2$$

$$Z = 2X - 5Y$$

$$E(Z) = E(2X - 5Y)$$

$$= E(2X) - E(5Y)$$

$$= 2 \cdot E(X) - 5 \cdot E(Y)$$

$$= 2(2) - 5(3)$$

$$= 4 - 15$$

$$= \boxed{-11}$$

$$\sigma_z^2 = \sigma_{(2X - 5Y)}^2$$

$$= \sigma_{2X}^2 - \sigma_{5Y}^2$$

$$= (2)^2 \sigma_x^2 + (5^2) \sigma_y^2$$

$$= 4(1) + 25(2)$$

$$= 4 + 50$$

$$= \cancel{46} = \boxed{54}$$

$$= \cancel{46}$$

Q10.  $n = 600$

$$P(80 < X < 120)$$

$$= P(100 - 20 < X < 100 + 20)$$

$$K\sigma = 20$$

$$K = \frac{20}{\sqrt{500/6}}$$

$$\mu = 100$$

$$\sigma = 120 / 80$$

$$K = \frac{|X - \mu|}{\sigma}$$

$$K\sigma = X - \mu = 20$$

Lower bound :

$$P\{|X - \mu| < K\sigma\} \geq 1 - \frac{1}{K^2}$$

$$E(X) = np = \frac{1}{6} \times 600 = 100, \quad p = \frac{1}{6}, \quad q = \frac{5}{6}, \quad n = 600$$

$$\text{Var}(X) = npq = \frac{1}{6} \times \frac{5}{6} \times 600 = \frac{500}{6}$$

$$P\{|X - 100| < K\sqrt{500/6}\} \geq 1 - \frac{1}{K^2}$$

$$P\{|X - 100| < 20\} \geq 1 - \frac{1}{\left\{\frac{20}{\sqrt{500/6}}\right\}^2} = 1 - \frac{5}{24}$$

$$= \boxed{\frac{19}{24}}$$

$$Q11. \quad F(x) = \begin{cases} 0, & x < 0 \\ \frac{x}{2}, & 0 \leq x < 1 \\ \frac{1}{2}, & 1 \leq x < 2 \\ \frac{x}{4}, & 2 \leq x < 4 \\ 1, & x \geq 4 \end{cases}$$

$$f(x) = \frac{d}{dx} F(x)$$

$$\therefore f(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{2}, & 0 \leq x < 1 \\ 0, & 1 \leq x < 2 \\ \frac{1}{4}, & 2 \leq x < 4 \\ 0, & x \geq 4 \end{cases}$$

Probability  
Density  
Function (pdf)

$$Q12. \quad f(x) = \begin{cases} \frac{100}{x^2}, & x \geq 100 \\ 0, & \text{elsewhere} \end{cases}$$

$$(i) \quad P(X < 150) = \int_{100}^{150} \frac{100}{x^2} \cdot dx$$

$$= 100 \left[ \frac{-1}{x} \right]_{100}^{150}$$

$$= -100 \left[ \frac{1}{150} - \frac{1}{100} \right]$$

$$= \frac{-100}{50} \left[ \frac{1}{3} - \frac{1}{2} \right] = -2 \left[ \frac{2-3}{6} \right] = -2 \left( \frac{-1}{6} \right)$$

$$= \frac{1}{3}$$

All three  
 $= \left( \frac{1}{3} \right) \times \left( \frac{1}{3} \right) \times \left( \frac{1}{3} \right)$

$$= \boxed{\frac{1}{27}}$$

$$(ii) \quad P(X > 150) = 1 - P(X < 150)$$

$$= 1 - \frac{1}{3}$$

$$= \frac{2}{3}$$

For all 3 tubes

$$= \left( \frac{2}{3} \right)^3$$

$$= \boxed{\frac{8}{27}}$$

$$(iii) \quad P(150 \leq X < 200) = \int_{150}^{200} \frac{100}{x^2} \cdot dx$$

$$= -100 \left[ \frac{1}{x} \right]_{150}^{200}$$

$$= -100 \left[ \frac{1}{200} - \frac{1}{150} \right] = \frac{-100}{50} \left[ \frac{1}{4} - \frac{1}{3} \right] = -2 \left[ \frac{3-4}{12} \right]$$

$$= \frac{1}{6}$$

$$\begin{aligned}
 & P(X < 200 \mid X \geq 150) \\
 &= \frac{P(X < 200) \cap P(X \geq 150)}{P(X \geq 150)} \\
 &= \frac{P(150 \leq X < 200)}{P(X \geq 150)} \\
 &= \frac{1}{6} \times \frac{3}{2} \\
 &= \boxed{\frac{1}{4}}
 \end{aligned}$$

(iv)  $P(X \geq 150)$  for 1 tube =  $\frac{2}{3}$   
 For  $n$  tubes,

$$\left(\frac{2}{3}\right)^n = 0.5$$

$$n \ln\left(\frac{2}{3}\right) = \ln(0.5)$$

$$n = \frac{\ln(0.5)}{\ln\left(\frac{2}{3}\right)} = \frac{-0.6931}{-0.4055} \approx 1.71$$

$$\therefore \boxed{n_{\max} = 1}$$

Q13.  $E(X) = 4 = 2$   
 $E(X^2) = 1 \cdot \left(\frac{1}{2}\right) + 4\left(\frac{1}{4}\right) + 9\left(\frac{1}{8}\right) + \dots$

Let  $E(X^2) = S$

then,  $\frac{S}{2} = \frac{1}{4} + 4\left(\frac{1}{8}\right) + 9\left(\frac{1}{16}\right) + \dots$

$S - \frac{S}{2} = \frac{1}{2} + \frac{3}{4} + \frac{5}{8} + \frac{7}{16} + \dots$

$\hookrightarrow$  A.G.P

Sum of A.G.P. =  $\frac{a}{1-r} + \frac{dr}{(1-r)^2}$   
 $= \frac{1}{1-1/2} + \frac{2(1/2)}{(1-1/2)^2}$   
 $= \frac{1}{1/2} + \frac{1}{(1/2)^2} = 6$

$\therefore \text{Var}(X) = E(X^2) - \{E(X)\}^2$   
 $= 6 - (2)^2$   
 $= 2$

$K\sigma = C \quad \therefore K = \frac{2}{\sqrt{2}} = \sqrt{2}$

$P\{|X-2| \leq 2\} \geq 1 - \frac{1}{K^2}$

$P\{-2 \leq X-2 \leq 2\} \geq 1 - \frac{1}{2} = \frac{1}{2}$

$\therefore \boxed{P\{0 \leq X \leq 4\} \geq \frac{1}{2}}$

~~Therefore,~~

$P\{X = 0, 1, 2, 3, 4\}$

$= P(0) + P(1) + P(2) + P(3) + P(4)$

$= 0 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$

$= \boxed{\frac{15}{16}}$

Hence, proved.

Q15. (i) Let  $x$  = no. of trials to open door

$$p = \frac{1}{n}$$

$$q = 1 - \frac{1}{n} = \frac{n-1}{n}$$

$$E(x) = \frac{1}{p} = \frac{1}{1/n} = n$$

$$\text{Var}(x) = \frac{1-p}{p^2} = \frac{1 - \frac{1}{n}}{1/n^2} = \frac{n-1}{1/n} = n^2 - n$$

$$(ii) P(x=1) = \frac{1}{n}$$

$$P(x=2) = \frac{n-1}{n} \left( \frac{1}{n-1} \right) = \frac{1}{n}$$

$$P(x=3) = \frac{n-1}{n} \left( \frac{n-2}{n-1} \right) \left( \frac{1}{n-2} \right) = \frac{1}{n}$$

$$P(x=n) = \frac{1}{n}$$

$$\therefore E(x) = 1 \cdot \frac{1}{n} + 2 \cdot \frac{1}{n} + \dots = \frac{1}{n} \left( \frac{n(n+1)}{2} \right)$$

$$E(x) = \frac{n+1}{2}$$

$$E(x^2) = 1 \cdot \frac{1}{n} + 4 \cdot \frac{1}{n} + 9 \cdot \frac{1}{n} + \dots$$

$$= \frac{1}{n} (1+4+9+\dots) = \frac{1}{n} \left( \frac{n(2n+1)(n+1)}{6} \right)$$

$$= \frac{(2n+1)(n+1)}{6}$$

$$\text{Var}(x) = E(x^2) - [E(x)]^2$$

$$= \frac{(2n+1)(n+1)}{6} - \left( \frac{n+1}{2} \right)^2$$

$$= \frac{n^2-1}{12}$$

Q16. Probability distribution of  $y$

| $x$ | $P(x)$ | $y = x^2 + 2x$ | $P(y)$ |
|-----|--------|----------------|--------|
| 1   | 0.1    | $1+2 = 3$      | 0.1    |
| 2   | 0.2    | $4+4 = 8$      | 0.2    |
| 3   | 0.5    | $9+6 = 15$     | 0.5    |
| 4   | 0.2    | $16+8 = 24$    | 0.2    |

Cumulative distribution of  $y$

| $y$ | $F(y)$ |
|-----|--------|
| 3   | 0.1    |
| 8   | 0.3    |
| 15  | 0.8    |
| 24  | 1      |

Mean of  $y$  :

$$\begin{aligned} E(y) &= \sum y \cdot P(y) \\ &= 3(0.1) + 8(0.2) + 15(0.5) + 24(0.1) \\ &= 11.8 \end{aligned}$$

Variance of  $y$  :

$$\begin{aligned} E(y^2) &= \sum y^2 (P(y)) \\ &= 9(0.1) + 64(0.2) + 225(0.5) + 576(0.1) \\ &= 183.8 \end{aligned}$$

$$\begin{aligned} \text{Var}(y) &= E(y^2) - [E(y)]^2 \\ &= 44.56 \end{aligned}$$